BioOne RESEARCH EVOLVED

Adjuvant Effects of Chitosan and Calcium Phosphate Particles in an Inactivated Newcastle Disease Vaccine

Author(s): Marina A. Volkova , Anna V. Irza , Irina A. Chvala , Sergy F. Frolov , Vladimir V. Drygin , and Darrell R. Kapczynski

Source: Avian Diseases, 58(1):46-52. 2014.
Published By: American Association of Avian Pathologists
DOI: http://dx.doi.org/10.1637/10510-020413-Reg.1
URL: http://www.bioone.org/doi/full/10.1637/10510-020413-Reg.1

BioOne (www.bioone.org) is a nonprofit, online aggregation of core research in the biological, ecological, and environmental sciences. BioOne provides a sustainable online platform for over 170 journals and books published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Web site, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/page/terms_of_use.

Usage of BioOne content is strictly limited to personal, educational, and non-commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

AVIAN DISEASES 58:46–52, 2014

Adjuvant Effects of Chitosan and Calcium Phosphate Particles in an Inactivated Newcastle Disease Vaccine

Marina A. Volkova,[A] Anna V. Irza,[A] Irina A. Chvala,[A] Sergy F. Frolov,[A] Vladimir V. Drygin,[A] and Darrell R. Kapczynski[BC]

[A]Russian Research Institute for Animal Health, FGI Federal Centre for Animal Health, Vladimir, Russia 600901
[B]Exotic and Emerging Avian Disease Research Unit, Southeast Poultry Research Laboratory, Agricultural Research Service, US Department of Agriculture, 934 College Station Road, Athens, GA 30605

Received 11 February 2013; Accepted 25 September 2013; Published ahead of print 30 September 2013

SUMMARY. The adjuvant activity of chitosan (CS) and calcium phosphate (CAP) particles was studied following intranasal (mucosal) administration to commercial chickens with inactivated Newcastle disease virus (NDV) vaccine. After three vaccinations with inactivated NDV in combination with CS or CAP an increase in antibody titers in blood and mucosal samples in chickens was observed when compared with the administration of NDV antigen only. A lower level of humoral immunity was observed in broiler chickens compared to layer-type birds. The CS-based vaccine demonstrated higher antigenic and protective activity following lethal challenge than the vaccine containing CAP. Because CS particles efficiently changed mucosal and humoral immunity and protective activity, CS may in the future be considered for use as a potential adjuvant for production of vaccines for poultry.

RESUMEN. Efectos adyuvantes del quitosano y de partículas de fosfato de calcio en una vacuna inactivada contra la enfermedad de Newcastle.

Se estudió la actividad adyuvante del quitosano (CS) y de partículas de fosfato de calcio (CAP) después de una administración intranasal (mucosa) en pollos comerciales con una vacuna inactivada contra el virus de la enfermedad de Newcastle. Después de tres vacunaciones con la vacuna inactivada en combinación con el quitosano o con el fosfato de calcio, se observó un aumento en los títulos de anticuerpos en las muestras de sangre y de las mucosas de los pollos en comparación con la administración del antígeno de Newcastle solo. Se observó un nivel más bajo de inmunidad humoral en los pollos de engorde en comparación con las aves de postura. La vacuna con quitosano demostró una mayor actividad antigénica y protectora después de la exposición letal en comparación con la vacuna que contenía fosfato de calcio. Debido a que las partículas de quitosano cambiaron de manera eficiente inmunidad de las mucosas, la inmunidad humoral, además de la actividad protectora, el quitosano puede ser considerado en el futuro para su uso como un potencial adyuvante en la producción de vacunas en avicultura.

Key words: Newcastle disease, inactivated vaccine, adjuvant, chitosan, calcium phosphate particles

Abbreviations: CAP = calcium phosphate; CS = chitosan; EID_{50} = 50% embryo-infectious dose; HI = hemagglutination inhibition; Ig = immunoglobulin; ND = Newcastle disease; NDV = Newcastle disease virus; PBS = phosphate-buffered saline; RRT-PCR = real-time reverse transcriptase–PCR; T = thymus-derived; TPP = triphosphate pentabasic

For the majority of virus pathogens that affect poultry, surfaces of the mucous membranes of respiratory and gastrointestinal tracts are the main entry points of infection. In addition, the first contact of immune cells with an antigen occurs on these epithelial surfaces. It is known that at early stages of infection the immune response is expressed mainly by innate, nonspecific immune factors, including cytokines and interferons. During the course of infection a specific immune response is created when antibodies and thymus-derived (T) cells target epitopes found on viral proteins (1,7,20,21,26). It has long been known that induction of local immunity, including immunoglobulin (Ig) IgA antibodies, on mucous membranes can play an important part in controlling infection by these viruses. Vaccination by specific vaccine antigens via mucous membranes possesses the immunologic capability for prevention of infection through mucous membranes (1,2,7). Mucosal vaccines containing specific antigens and natural or synthetic adjuvants with mucoadhesive properties present an upcoming trend in terms of efficacy, increase in local immunity, and facilitation of administration of immunologic preparations.

Vaccine adjuvants are chemical substances, microbial components, or proteins that enhance an immune response after the administration of inactivated vaccines (4,9,22). The choice of an adjuvant is an important factor for designing mucosal vaccines. An adjuvant for mucosal vaccines should be capable of surviving in respiratory and gastrointestinal tracts and enhancing antigen delivery to immune cells. Both chitosan (CS) and calcium phosphate (CAP) particles have been previously tested as mucosal adjuvants with positive results (3,5,8,11,17,19,23). Mucosal vaccines have additional properties as compared with conventional vaccines in that they induce both local protection of mucous membranes and systemic immunity (3,6,15,17,18,24). The efficacy of mucosal vaccines can be evaluated under experimental conditions using serologic methods and protection against challenge.

These studies were aimed at studying the mucosal and humoral immune response of chickens to intranasal administration of Newcastle disease (ND) vaccine preparations containing CAP particles or CS as adjuvants. Protective immunity against challenge with a highly virulent ND virus (NDV) was determined at different intervals following immunization.

MATERIALS AND METHODS

Viruses. The lentogenic NDV vaccine strain LaSota was utilized during this study. The velogenic vicerotropic strain Chicken/Russia/Amursky/1057/2006 (Ck/Russia/06) was used as challenge virus. This isolate has an intracranial pathogenicity index of 2.0. NDV was

[C]Corresponding author. E-mail: darrell.kapczynski@ars.usda.gov

propagated and titrated in 9- to 11-day-old specific-pathogen-free chicken embryos via the chorioallantoic sac route. Infectious titer of virus was approximately 10^8 50% embryo-infectious dose (EID_{50}) per milliliter.

Vaccines. The purification and concentration of aminoethyl ethyleneimine–inactivated virus were carried out using a method that included low-speed centrifugation and ultracentrifugation at 100,000 × *g* for 1.5 hr at 4 C. Titer of the NDV concentrated antigen determined by hemagglutination test was 15 $\log_2$.

CS microparticles were prepared according to the procedures described by A. Vila *et al.* based on the ionotropic gelation of CS (Sigma, St. Louis, MO) with sodium triphosphate pentabasic (TPP) (Sigma) anions (23). The particles are formed spontaneously upon addition of 2.4 ml of 15% aqueous TPP solution to 6 ml of CS solution (2 mg/ml), under magnetic stirring. For association of NDV with CS microparticles, antigen was incorporated in the TPP solution (90 ml of antigen). Microparticles were isolated by centrifugation at 100,000 × *g* for 1 hr on a 30% sucrose buffer. Supernatants were discarded and microparticles were resuspended in 90 ml of phosphate-buffered saline (PBS) and stirred overnight at 4 C using a magnetic stirrer.

The vaccine with CAP nanoparticles was prepared in two stages according to the procedures described by He *et al.* (8). First, 4 ml NDV antigen was added to 30 ml 12.5 mM calcium chloride, followed by the addition of 30 ml 12.5 mM dibasic sodium phosphate and 6 ml of 15.6 mM sodium citrate. At the second stage the coincubation of produced particles with 62 ml of NDV antigen was carried out. Both procedures took place with constant stirring for 1 hr at 4 C using a magnetic stirrer. The control vaccines were NDV antigen plus PBS (consisting of equal volumes) and PBS alone. Formulation of NDV vaccines with CS or CAP retained approximately 89%–92% antigen protein.

Vaccination Experiment 1. Twenty 3-wk-old broilers from a commercial poultry facility were divided into four groups of five birds. Birds in groups 1 and 2 received 200 μl of mucosal vaccine with CAP or CS, respectively. Birds in groups 3 and 4 received 200 μl of NDV in PBS or PBS alone, respectively. All groups of chickens were vaccinated intranasally three times with an interval of 5 days. Serum and mucosal samples (tracheal swabs and lacrimal fluid) were taken at 0, 1, 2, 3, and 4 wk postvaccination. Four weeks postvaccination birds in all groups were challenged through an intramuscular route with $10^{6.0}$ EID_{50} per bird with Ck/Russia/06 in a volume of 0.5 ml. Birds were monitored daily for morbidity and mortality. Ten days postchallenge all surviving birds were euthanatized by approved procedures and serum samples and spleens were taken.

Vaccination Experiment 2. One hundred 3-wk-old layer chickens from commercial layer poultry were divided into four groups of 25 birds. Birds in groups 1 and 2 received 200 μl of mucosal vaccine mixed with CAP or CS, respectively. Birds in groups 3 and 4 received 200 μl of NDV in PBS or PBS alone, respectively. All groups of chickens were vaccinated intranasally three times with an interval of 5 days. Serum and mucosal samples (tracheal swabs and lacrimal fluid) were taken at 0, 1, 2, 3, 4, and 8 wk post-vaccination. At 1, 2, 3, 4, and 8 wk postvaccination birds in all groups were challenged via an intramuscular route with $10^{6.0}$ EID_{50} per bird with Ck/Russia/06 in a volume of 0.5 ml. Birds were monitored daily for morbidity and mortality, with cloacal swabs taken on days 3 and 7 postchallenge for virus detection (13).

ELISA. NDV-specific antibodies were detected by an end-point dilution ELISA (10). Briefly, 96-well microtiter plates (Nunc-Immuno™, Sigma) coated with NDV antigen (1 μg/ml) were incubated with serial dilutions of samples of serum, tracheal swabs, lacrimal fluids, or controls for 30 min at 37 C. The plates were washed with wash buffer and incubated with horseradish peroxidase–conjugated anti-chicken Ig A or IgM (Serotec, Kidlington, UK) or IgG (Synbiotics, Kansas City, MO) for 30 min at 37 C, followed by washing with wash buffer and development with 2,2′-azino-bis(3-ethylbenzothiazoline 6-sulphonic acid) (Synbiotics) at room temperature until color developed. The reaction was stopped using 1% sodium dodecyl sulfate. The result was recorded as optical density at 405 nm. Titers for IgA, IgM, and IgG in serum and mucosal samples were expressed as the group mean ± standard error of the mean for that group.

Hemagglutination inhibition (HI) assay. HI tests were performed by standard microtiter plate method (16). The HI tests were performed by use of a commercial kit from FGI ARRIAH (Vladimir, Russia), with 4 hemagglutinin units per well. HI titers determined to be 4 ($\log_2$) or higher were regarded as positive.

Flow cytometric analysis. Spleens ($n = 3$ per group) were removed for cytometric analysis and immediately placed on ice (12). Each organ was passed through a fine mesh sieve to obtain a single-cell suspension. Spleen cells were then overlaid on Ficoll-Paque, density 1.077 g/ml (Amersham Biosciences, Uppsala, Sweden), and centrifuged for 30 min at 400 × *g* at 8 C. The spleen cells containing interface were transferred to new tubes and washed twice with PBS by centrifugation at 300 × *g* for 10 min. Fifty microliters of the spleen cells with a density of 1×10^6 cells/ml were incubated with 2 μl anti-chicken CD4-FITC and CD8α-FITC (Southern Biotech, Birmingham, AL) monoclonal antibodies for 30 min at 8 C. Positively stained cells were washed with PBS followed by another centrifugation step and were analyzed by BD FACS™ Calibur (Becton Dickinson, Franklin Lakes, NJ) using Cell Quest Pro 1.0 software. A minimum of 10,000 events were collected for each sample.

Real-time reverse transcriptase–PCR (RRT-PCR). Detection of NDV in biologic materials was obtained using RRT-PCR as described with minor modifications (25). RNA was extracted with use of the robot-based station Freedom EVO-100 (Tecan, Männedorf, Switzerland) from 0.1 ml of cloacal swab material using a nucleic acid extraction kit with magnetic particles (Magna DNA Prep 100, Roche, Indianapolis, IN) according to the manufacturer's instructions. Specific primers (Am-BR-NDV-F72 5′-CTCACTCCTCTTGGCGACTC-3′ and Am-BR-NDV-R246 5′-TGGCAGCATTCTGGTTGG-3′) and fluorescently labeled oligonucleotide TaqMan probe (Am-BR-NDV-207 5′-FAM–CTGTTATCTGTGCCGCTGTTGC–BHQ1-3′) were designed for the NDV challenge isolate (Ck/Russia/06) F gene segment. RRT-PCR was performed in a programmed real-time DNA amplifier (Rotogene 6000, Qiagen, Valencia, CA) using positive and negative controls containing NDV RNA and no template, respectively.

Statistical analysis. The nonparametric Mann-Whitney *U*-test was used to determine statistical differences between the vaccinate group and the control group using the software package Statistica 6.0 (StatSoft, Moscow, Russia). Differences were considered as significance at $P < 0.05$.

RESULTS

Experiment 1. The effect of CAP particles and CS-based particles on the immune response of broiler chicks to NDV was studied.

IgA antibodies were detected in both tracheal swabs and lacrimal fluids from NDV-vaccinated chicks at 1 wk following the third vaccination (Figs. 1, 2). The maximum production of IgA in tracheal swabs was registered at 1 wk following vaccination. Although significant differences in IgA titer were observed in the NDV with CS and NDV in PBS groups, no significant increase in IgA was observed with NDV with CAP (Fig. 1). In lacrimal fluids significant increases in the amount of IgA were detected in all groups compared to the control birds at all days tested (Fig. 2). Statistically significant increased IgM and IgG values were recorded in tracheal and lacrimal swabs in all groups receiving NDV for almost all time points. In lacrimal fluids the maximum amount of IgM was observed at 1 wk following vaccination and the maximum amount of IgG was detected at 3 and 4 wk following vaccination of chicks by CS and CAP particles, respectively. IgA and IgG concentrations in tracheal swabs at 4 wk following the third vaccination were higher in groups of chicks immunized by mucosal preparations than in chicks immunized by only inactivated antigen.

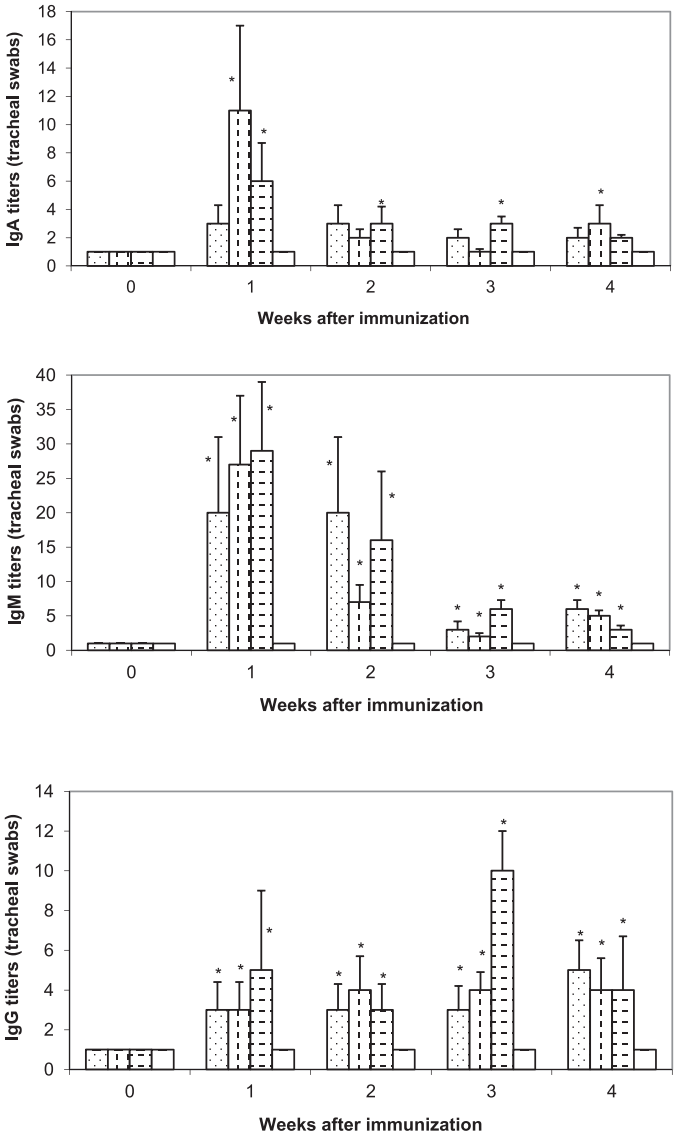

Fig. 1. Tracheal swab antibody response in Experiment 1 following intranasal vaccination of commercial broilers (at 20 days of age) with experimental mucosal inactivated virus vaccines. Each bar represents the group mean antibody level (+SEM) for chicken immunized with NDV + CAP (points), NDV + CS (vertically striped bars), NDV + PBS (horizontally striped bars), and controls (open bar). *Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

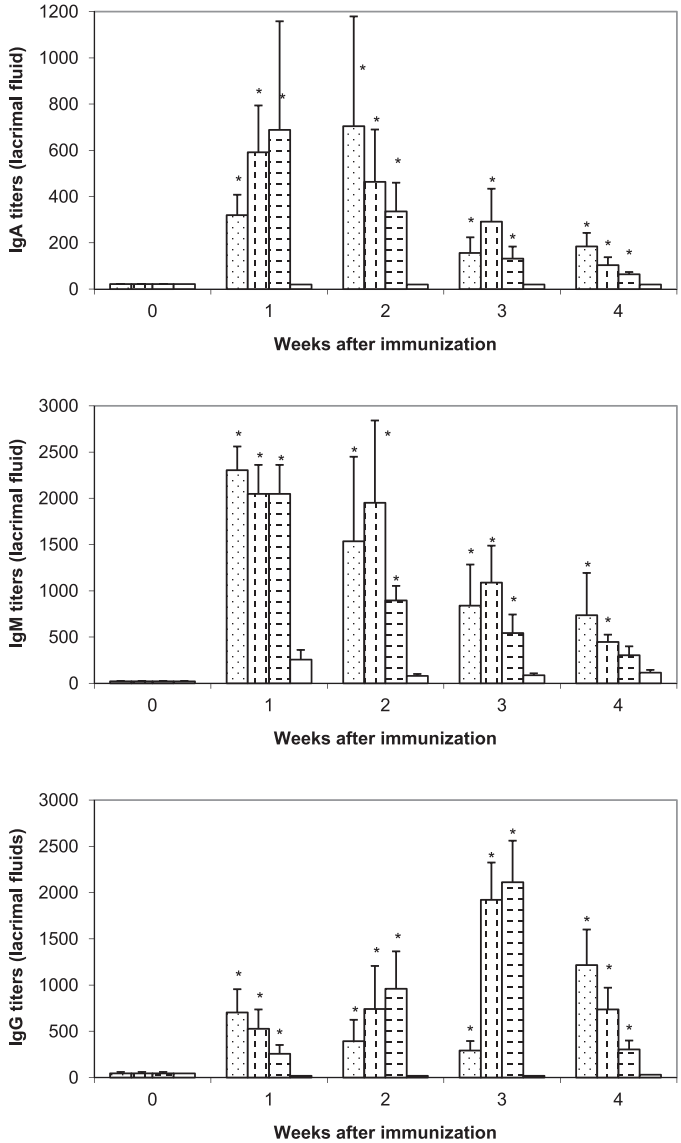

Fig. 2. The lacrimal fluid antibody response in Experiment 1 following intranasal vaccination of commercial broilers (at 20 days of age) with experimental mucosal inactivated-virus vaccines. Each bar represents the group mean antibody level for chickens immunized with NDV + CAP (points), NDV + CS (vertically striped bars), NDV + PBS (horizontally striped bars), and controls (open bar). *Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

Overall, the humoral antibody response to the administration of experimental NDV vaccines was low, although some statistically increased antibody levels were observed, predominantly at 1 wk postvaccination (Table 1). A short-term increase in the amount of IgM in the blood of chicks immunized by experimental vaccines was observed following vaccination (Table 1). The HI results demonstrated the maximum number of positive samples from chicks that were vaccinated with CAP particles at 1 wk following the third inoculation (Table 1).

The challenge of chicks by highly virulent NDV at 4 wk following the third vaccination was conducted to determine protective properties of experimental vaccines (Table 2). The obtained results demonstrated that the ND vaccine with CAP particles created the highest level of protection in chicks against infection by virulent NDV. Nonspecific signs of the disease occurred 1–8 days following challenge in one out of three chicks; later they disappeared and chicks became clinically healthy. In the group of chicks inoculated by the vaccine with CS, ND-specific signs appeared in one out of three chicks on the second day following challenge and that chick died. In the group of chicks vaccinated by inactivated antigen without an adjuvant, specific signs of the disease were observed in two out of two chicks and one chick died.

The immunophenotyping and quantitative evaluation of T-lymphocyte subpopulations in spleens at 4 wk following the third vaccination and at 10 days following challenge were conducted using flow cytometry to determine cellular immune stimulation after application of different mucosal adjuvants. Results of quantitative determination of T-lymphocyte subpopulations are presented in

Table 1. Serum antibody response in Experiment 1 following intranasal vaccination of commercial broilers (at 20 days of age) with experimental mucosal inactivated-virus vaccines and challenge with Ck/Rus/06.

Group	Test	Postvaccination sample					Postchallenge sample
		Week 0	Week 1	Week 2	Week 3	Week 4	
NDV + CAP	HI[A]	3.9 (1.4)[B]	4.3 (2.2)*	3.6 (1.0)*	2.0 (1.1)	3.3 (1.2)	14.0 (0.2)
		6/10[C]	4/5	1/5	0/5	2/5	
	ELISA IgM	50 (0)[B]	360 (116)*	920 (293)*	1880 (1155)*	280 (132)	NT[D]
	ELISA IgG	465 (79)[B]	300 (126)*	100 (27)	340 (125)*	360 (117)	NT
NDV + CS	HI	3.9 (1.4)	2.8 (1.0)*	2.3 (1.3)	1.0 (0.8)	1.0 (0.5)	14.0 (0.5)
		6/10	1/5	1/5	0/5	0/5	
	ELISA IgM	50 (0)	120 (20)	680 (250)*	440 (98)	180 (20)	NT
	ELISA IgG	465 (79)	200 (55)*	250 (140)*	370 (126)	240 (40)	NT
NDV + PBS	HI	3.9 (1.4)	2.0 (1.0)	1.0 (0)	0	1.0 (0)	14.0 (0.2)
		6/10	0/5	0/5	0/5	0/5	
	ELISA IgM	50 (0)	240 (68)*	280 (49)	520 (120)	140 (24)	NT
	ELISA IgG	465 (79)	70 (12)	120 (34)	60 (10)	200 (55)	NT
Controls	HI	3.9 (1.4)	0	0	0	0	NT
		6/10	0/5	0/5	0/5	0/5	
	ELISA IgM	50 (0)	50 (0)	160 (37)	180 (64)	140 (66)	NT
	ELISA IgG	465 (79)	50 (0)	50 (0)	50 (0)	70 (12)	NT

[A]Mean titer expressed as reciprocal $\log_2$. Titers ≥ 4 are considered positive.
[B]Results are expressed as mean, with standard error of the mean in parentheses.
[C]Positive samples/total number of samples.
[D]NT = not tested.
*Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

Table 3. In groups of chicks immunized with experimental vaccines, $CD4^+$ subpopulations were greatest in birds receiving only antigen (NDV in PBS) prior to challenge. Following challenge, both the CS and PBS groups decreased the percentage of $CD4^+$-positive cells in the spleen. Interestingly, the highest $CD4^+$ levels were observed in groups receiving adjuvant (CS or CAP). The highest percentage of $CD8^+$ subpopulations in spleens following NDV vaccination was observed in birds receiving the mucosal adjuvants. Following challenge, all groups of birds demonstrated decreases in $CD8^+$ populations, although higher levels were observed in the groups receiving mucosal adjuvants.

Experiment 2. The experiment was aimed at studying immunobiologic properties of mucosal NDV vaccines with CS and CAP in commercial egg-laying chicks. Results of testing blood sera by HI are given in Table 4. Significantly significant increases in HI titers were observed in all groups receiving NDV. In chicks immunized with mucosal adjuvant vaccine preparations, the maximum amount of specific antibodies was detected at 2 wk following vaccination. In contrast, the maximum level of NDV antibodies immunized by antigen in PBS was observed at 4 wk postvaccination. Overall, the number of positive samples was higher in chicks vaccinated with CS as compared to those vaccinated with CAP particles.

Levels of IgA, IgM, and IgG in tracheal swabs and lacrimal fluids were determined by ELISA (Figs. 3, 4). Almost all vaccinated birds demonstrated significantly higher IgA, IgM, and IgG antibodies in tracheal swabs compared to control birds at all time points tested. The maximum amount of antibodies was detected in tracheal swabs at 2 wk and in lacrimal fluids at 4 wk following the last vaccination. Only a few groups demonstrated significantly higher IgA levels in lacrimal fluid, with all NDV groups higher at 4 wk. However, significantly higher IgM and IgG titers were demonstrated in the NDV-vaccinated groups at most time points tested. At 8 wk following vaccination no IgA antibodies were detected in mucosal samples and the IgG levels were higher in chicks receiving CS.

Results of ELISA testing also demonstrated that the maximum amount of IgM antibodies was observed at 2 wk following the third vaccination in all vaccinated groups and the maximum amount of IgG antibodies was detected at 2 wk in groups immunized by vaccine with CS, and at 4 weeks in birds inoculated by vaccine with CAP and NDV in PBS (Fig. 3). The highest levels of serum IgM and IgG antibodies were observed in chicks immunized by vaccine with CS as an adjuvant.

Chicks were challenged by highly virulent NDV at different dates following vaccination for detection of protective properties of experimental vaccines (Table 5). The challenge of chicks at 7 days following immunization resulted in death of all chicks in the control group and in the group immunized by antigen without an adjuvant (NDV in PBS). In the immunized group of chicks with mucosal adjuvants (CAP and CS), two out five chicks in each group died with specific clinical signs attributive to ND. Mortality of numerous birds in vaccinated groups was also observed when birds were challenged at 14 days following immunization. The full protection of chicks

Table 2. Protective activity of mucosal inactivated-virus vaccines following lethal challenge in Experiment 1.

Group	Clinical signs of NDV					Mortality				
	2[A]	4	6	8	10	2	4	6	8	10
NDV + CAP	1/3[B]	1/3	1/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
NDV + CS	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3
NDV + PBS	1/2	2/2	2/2	2/2	2/2	0/2	1/2	1/2	1/2	1/2
Controls	2/2	2/2	2/2	2/2	2/2	1/2	2/2	2/2	2/2	2/2

[A]Days after challenge.
[B]Number of chickens with the demonstration of definite effect/total number of chickens.

Table 3. Flow cytometric analysis of the effect of mucosal vaccination and NDV infection on the relative proportion of $CD4^+$ and $CD8^+$ in splenic samples ($n = 3$).

Group	$CD4^+$		$CD8^+$	
	Postvaccination sample	Postchallenge sample	Postvaccination sample	Postchallenge sample
NDV + CAP	10.92 ± 0.13[A]	13.56 ± 0.46	55.28 ± 0.30	35.5 ± 1.28
NDV + CS	14.75 ± 0.41	11.83 ± 0.72	52.27 ± 0.45	39.68 ± 2.33
NDV + PBS	16.57 ± 0.19	8.83 ± 0.3	45.75 ± 0.48	32.36 ± 0.85
Controls	13.56 ± 0.50	NT[B]	45.49 ± 0.18	NT

[A]Results expressed as percentage of positive samples.
[B]NT = not tested.

against virulent virus was registered in all experimental groups when chicks were challenged on day 21 following vaccination. Chicks from the group immunized by vaccine with CS were resistant to virulent NDV on day 28 following immunization. In groups of chicks immunized with CAP or NDV antigen in PBS, mortality of one and two chicks, respectively, was observed. At challenge 56 days postimmunization, one bird in each of the CS and NDV-in-PBS groups, and two birds in the CAP group, died after challenge.

RRT-PCR was used for the evaluation of NDV shedding in vaccinated chicks following challenge by a highly virulent NDV. Cloacal swabs from control and vaccinated chicks at 7, 14, and 21 days following the third immunization were tested as shown in Table 6. The frequency of NDV-positive swabs in the vaccinated groups appeared to decline from day 3 to day 7. In addition, fewer positive swabs were observed on day 3 postchallenge when birds were challenged at 21 days after immunization as compared to earlier challenge points (e.g., days 7 and 14).

DISCUSSION

The effect of CAP- and CS-based particles as mucosal adjuvants in inactivated NDV vaccines was studied in chickens. The immunogenic and protective activity of mucosal vaccine preparations was tested using two serologic tests and challenge. Three types of biologic material samples (blood sera, tracheal swabs, and lacrimal fluids) were tested for a fuller characterization of immunogenicity of such vaccines.

The development of the mucosal immune response starts with the recognition of a pathogen by specialized cells of the epithelium that covers lymphoid follicles. It has been demonstrated that IgA is the primary antibody that mediates protection in mucous membranes, though immunoglobulins of M and G classes can also be detected (1,2,7). In our experiment, significantly higher IgA levels were detected in secretions from vaccinated chicks beginning from the first week following vaccination. Moreover, the maximum amount of IgA in broilers was observed at earlier date as compared with layer chicks.

The immunization by preparations with mucosal adjuvants caused the creation of significantly higher levels of humoral antibodies in layer chickens. The relatively low level of humoral immunity in broiler chicks after vaccination with CAP particles and CS could be explained by differences in the immune system between layers and broilers (14) and by the presence of maternal antibodies in chicks before vaccination.

Thus, the obtained results confirmed data of other authors that local administration of inactivated virus with mucosal adjuvant stimulated the local creation of antibodies in upper respiratory tracts and lacrimal fluids and the systemic induction of antibodies in blood sera.

The preliminary testing of protective immunity of mucosal vaccines in broiler chicks demonstrated their ability to induce

Table 4. Serum antibody response in Experiment 2 following intranasal vaccination of commercial chickens (at 20 days of age) with experimental mucosal inactivated-virus vaccines.

Group	Test	Postvaccination sample					
		Week 0	Week 1	Week 2	Week 3	Week 4	Week 8
NDV + CAP	HI[A]	2 (0.5)[B]	5.4 (0.3)*	4.1 (0.3)*	3.5 (0.4)*	3 (0.6)*	3.4 (0.6)*
		0/10[C]	9/10	7/10	4/10	2/5	1/5
	ELISA IgM	60 (7)	480 (137)*	700 (161)*	860 (270)	600 (126)	1040 (240)*
	ELISA IgG	215 (43)	75 (8)	900 (167)*	365 (152)*	1600 (438)*	280 (132)*
NDV + CS	HI	2 (0.5)	6 (0.4)*	5.1 (0.4)*	4.2 (0.5)*	3.9 (0.3)*	3.4 (0.6)*
		0/10	9/10	9/10	8/10	4/5	2/5
	ELISA IgM	60 (7)	540 (73)*	800 (186)*	600 (133)	440 (83)	1120 (196)*
	ELISA IgG	215 (43)	160 (44)	2120 (560)*	460 (60)*	1360 (122)*	520 (273)*
NDV + PBS	HI	2 (0.5)	4.5 (0.7)*	3.3 (0.5)*	2.6 (0.6)*	3.6 (0.9)*	1.7 (0.7)*
		0/10	8/10	6/10	3/8	3/5	0/5
	ELISA IgM	60 (7)	840 (268)*	500 (134)*	1950 (706)*	760 (240)	1200 (400)*
	ELISA IgG	215 (43)	110 (20)	450 (145)*	200 (46)*	1120 (528)*	70 (44)
Controls	HI	2 (0.5)	0	0	0	0	0
		0/10	0/10	0/10	0/10	0/5	0/5
	ELISA IgM	60 (7)	80 (8)	190 (28)	390 (57)	300 (63)	800 (0)
	ELISA IgG	215 (43)	50 (0)	50 (0)	50 (0)	100 (0)	63 (13)

[A]Mean titer expressed as reciprocal $\log_2$. Titers ≥4 are considered positive.
[B]Results are expressed as mean, with standard error of the mean in parentheses.
[C]Positive samples/total number of samples.
*Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

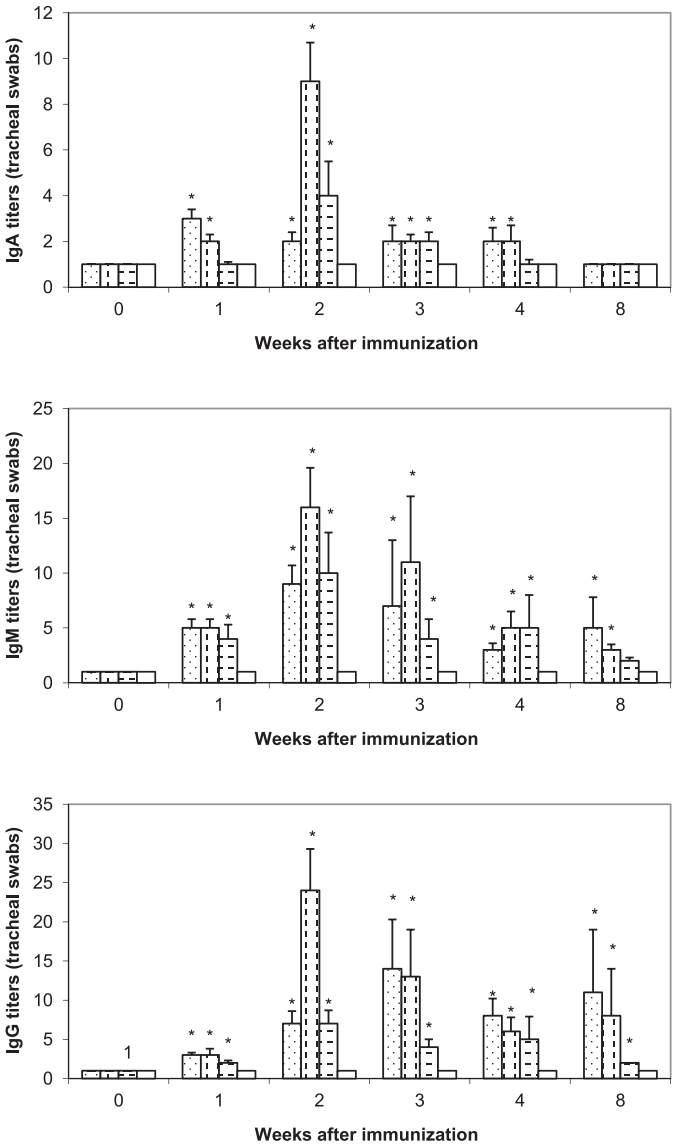

Fig. 3. Tracheal-swab antibody response in Experiment 2 following intranasal vaccination of commercial layers (at 20 days of age) with experimental mucosal inactivated-virus vaccines. Each bar represents the group mean antibody level (+SEM) for chickens immunized with NDV + CAP (points), NDV + CS (vertically striped bars), NDV + PBS (horizontally striped bars), and controls (open bar). *Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

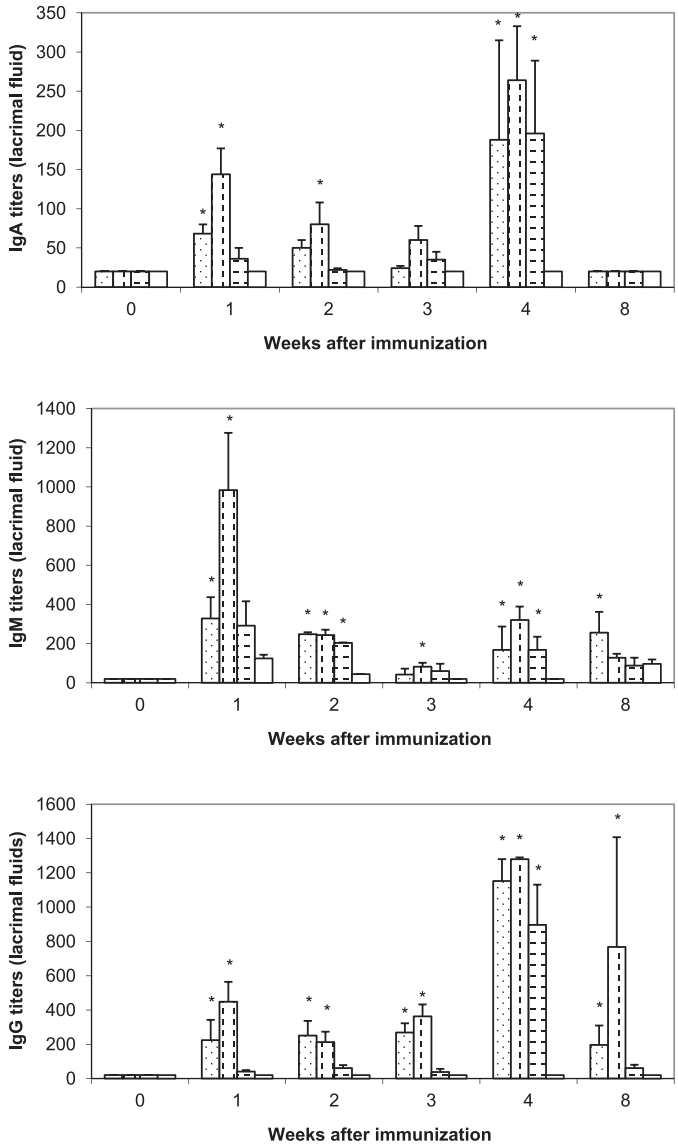

Fig. 4. The lacrimal fluid antibody response in Experiment 2 following intranasal vaccination of commercial layers (at 20 days of age) with experimental mucosal inactivated-virus vaccines. Each bar represents the group mean antibody level (+SEM) for chickens immunized with NDV + CAP (points), NDV + CS (vertically striped bars), NDV + PBS (horizontally striped bars), and controls (open bar). *Indicates titers that were confirmed to be significantly different from titers of the control group at $P < 0.05$; Mann-Whitney *U*-test.

protection against virulent NDV challenge. The CAP-particle vaccine protected chicks against both appearance of ND-specific signs and death, whereas the CS-based vaccine did not fully protected chicks against death. The protective immunity of mucosal vaccine preparations in layer chicks demonstrated specific protection against challenge by virulent NDV. The ND mucosal vaccine with CS demonstrated the highest level of protection. However, it should be noted that complete protective immunity was observed in all

Table 5. Protective activity of experimental mucosal inactivated virus vaccines following lethal challenge with Ck/Rus/06 in Experiment 2.

	Clinical signs of NDV					Mortality				
Group	7[A]	14	21	28	56	7	14	21	28	56
NDV + CAP	2/5[B]	1/5	1/5	0/5	2/5	2/5	2/5	0/5	1/5	2/5
NDV + CS	1/5	1/5	0/5	0/5	1/5	2/5	3/5	0/5	0/5	1/5
NDV + PBS	1/5	2/5	0/5	1/5	2/3	5/5	3/5	0/5	2/5	1/3
Controls	1/5	5/5	2/5	1/5	4/4	5/5	5/5	5/5	5/5	4/4

[A]Challenge at days postimmunization.
[B]Number of chickens with the demonstration of definite effects/total number of chickens.

Table 6. Frequency of cloacal swabs testing positive for NDV F gene by RRT-PCR (Experiment 2).

	Days postimmunization					
	7		14		21	
	Postchallenge sample day					
Group	Day 3	Day 7	Day 3	Day 7	Day 3	Day 7
NDV + CAP	2/5[A]	0/3	1/4	1/3	1/5	1/5
NDV + CS	1/5	0/3	1/5	0/2	0/5	0/5
NDV + PBS	2/5	0/1	2/5	0/2	0/5	0/5
Controls	1/1	—[B]	3/5	—	4/5	—

[A]Number positive/number tested.
[B]No survivors.

groups receiving NDV at various times after challenge, indicating induced immunity.

Due to the small number of experimental chicks used in these studies the evaluation of protective properties with mucosal adjuvants will be continued in the future with the aim of a more detailed characterization. Full-scale studies on the immune response of chicks to intranasal administration of experimental mucosal vaccines against ND demonstrated that the strongest humoral and mucosal responses were achieved following administration of vaccines with CS. The mucosal vaccine with CS also provided protection of chicks against the lethal ND at earlier dates as compared with other tested vaccine samples. Whether these types of vaccines are commercially viable remains to be determined, but inducing immunity at the site of viral infection remains an important consideration for future development.

REFERENCES

1. Al-Garib, S. O., A. L. Gielkens, E. Gruys, and G. Koch. Review of Newcastle disease virus with particular references to immunity and vaccination. World's Poult. Sci. J. 59:185–200. 2003.

2. Al-Garib, S. O., A. L. Gielkens, D. E. Gruys, L. Hartog, and G. Koch. Immunoglobulin class distribution of systemic and mucosal antibody responses to Newcastle disease in chickens. Avian Dis. 47:32–40. 2003.

3. Boonyo, W., H. E. Junginger, N. Waranuch, A. Polnok, and T. Pitaksuteepong. Chitosan and trimethyl chitosan chloride (TMC) as adjuvants for inducing immune responses to ovalbumin in mice following nasal administration. J. Control. Release 121:168–175. 2007.

4. Borges, O., G. Borchard, J. C. Verhoef, A. de Sousa, and H. E. Junginger. Preparation of coated nanoparticles for a new mucosal vaccine delivery system. Int. J. Pharm. 299:155–166. 2005.

5. Csaba, N., M. Koping-Hoggard, and M. J. Alonso. Ionically crosslinked chitosan/tripolyphosphate nanoparticles for oligonucleotide and plasmid DNA delivery. Int. J. Pharm. 382:205–214. 2009.

6. Dalgaard, T. S., L. R. Norup, A. R. Pedersen, K. J. Handberg, P. H. Jorgensen, and H. R. Juul-Madsen. Flow cytometric assessment of chicken T cell–mediated immune responses after Newcastle disease virus vaccination and challenge. Vaccine 28:4506–4514. 2010.

7. Hannant, D. Mucosal immunology: overview and potential in the veterinary species. Vet. Immunol. Immunopathol. 87:265–267. 2002.

8. He, Q., A. Mitchell, T. Morcol, and S. J. Bell. Calcium phosphate nanoparticles induce mucosal immunity and protection against herpes simplex virus type 2. Clin. Diagn. Lab. Immunol. 9:1021–1024. 2002.

9. Holmgren, J., C. Czerkinsky, K. Eriksson, and A. Mharandi. Mucosal immunisation and adjuvants: a brief overview of recent advances and challenges. Vaccine 21(Suppl. 2):S89–S95. 2003.

10. Hoshi, S., T. Nakamura, T. Nunoya, and S. Ueda. Induction of protective immunity in chickens orally immunized with inactivated infectious bursal disease virus. Vaccine 13:245–252. 1995.

11. Illum, L., I. Jabbal-Gill, M. Hinchcliffe, A. N. Fisher, and S. S. Davis. Chitosan as a novel nasal delivery system for vaccines. Adv. Drug Deliv. Rev. 51:81–96. 2001.

12. Kapczynski, D. R. Evaluating the cell-mediated immune response of avian species to avian influenza viruses. Methods Mol. Biol. 436:113–126. 2008.

13. Kapczynski, D. R., and D. J. King. Protection of chickens against overt clinical disease and determination of viral shedding following vaccination with commercially available Newcastle disease virus vaccines upon challenge with highly virulent virus from the California 2002 exotic Newcastle disease outbreak. Vaccine 23:3424–3433. 2005.

14. Koenen, M. E., A. G. Boonstra-Blom, and S. H. Jeurissen. Immunological differences between layer- and broiler-type chickens. Vet. Immunol. Immunopathol. 89:47–56. 2002.

15. Maas, R. A., M. Komen, M. van Diepen, H. L. Oei, and I. J. Claassen. Correlation of haemagglutinin-neuraminidase and fusion protein content with protective antibody response after immunisation with inactivated Newcastle disease vaccines. Vaccine 21:3137–3142. 2003.

16. [O.I.E.] World Organisation for Animal Health. Manual of diagnostic tests and vaccines for terrestrial animals: mammals, birds and bees, Biological Standards Commission. World Organisation for Animal Health, Paris. 2012.

17. Rauw, F., Y. Gardin, V. Palya, S. Anbari, M. Gonze, S. Lemaire, T. van den Berg, and B. Lambrecht. The positive adjuvant effect of chitosan on antigen-specific cell-mediated immunity after chickens' vaccination with live Newcastle disease vaccine. Vet. Immunol. Immunopathol. 134:249–258. 2010.

18. Rauw, F., Y. Gardin, V. Palya, S. van Borm, M. Gonze, S. Lemaire, T. van den Berg, and B. Lambrecht. Humoral, cell-mediated and mucosal immunity induced by oculo-nasal vaccination of one-day-old SPF and conventional layer chicks with two different live Newcastle disease vaccines. Vaccine 27:3631–3642. 2009.

19. Read, R. C., S. C. Naylor, C. W. Potter, J. Bond, I. Jabbal-Gill, A. Fisher, L. Illum, and R. Jennings. Effective nasal influenza vaccine delivery using chitosan. Vaccine 23:4367–4374. 2005.

20. Reynolds, D. L., and A. D. Maraqa. Protective immunity against Newcastle disease: the role of antibodies specific to Newcastle disease virus polypeptides. Avian Dis. 44:138–144. 2000.

21. Reynolds, D. L., and A. D. Maraqa. Protective immunity against Newcastle disease: the role of cell-mediated immunity. Avian Dis. 44:145–154. 2000.

22. Senel, S., and S. J. McClure. Potential applications of chitosan in veterinary medicine. Adv. Drug Deliv. Rev. 56:1467–1480. 2004.

23. Vila, A., A. Sanchez, K. Janes, I. Behrens, T. Kissel, J. L. Vila Jato, and M. J. Alonso. Low molecular weight chitosan nanoparticles as new carriers for nasal vaccine delivery in mice. Eur. J. Pharm. Biopharm. 57:123–131. 2004.

24. Wigley, P., and P. Kaiser. Avian cytokines in health and disease. Brazil. J. Poult. Sci. 3:1–8. 2003.

25. Wise, M. G., D. L. Suarez, B. S. Seal, J. C. Pedersen, D. A. Senne, D. J. King, D. R. Kapczynski, and E. Spackman. Development of a real-time reverse-transcription PCR for detection of newcastle disease virus RNA in clinical samples. J. Clin. Microbiol. 42:329–338. 2004.

26. Zekarias, B., A. A. Ter Huurne, W. J. Landman, J. M. Rebel, J. M. Pol, and E. Gruys. Immunological basis of differences in disease resistance in the chicken. Vet. Res. 33:109–125. 2002.

ACKNOWLEDGMENTS

This research was supported by the U.S. Department of Agriculture, Agricultural Research Service, OIRP Project 3460 and by the International Science and Technology Center, Moscow, Russia.